

INSTRUCTOR GUIDE

monil1_Introduction_to_Machine_Learning

LECTURE OVERVIEW

Main objectives

- Understand the fundamental concept of Machine Learning and its role within AI.
- Differentiate between the primary types of Machine Learning paradigms.
- Recognize the essential steps involved in a typical Machine Learning project.

Key learning outcomes

- Define Machine Learning and explain its distinction from Artificial Intelligence.
- Identify and provide examples of supervised, unsupervised, and reinforcement learning.
- Describe the basic workflow of developing and deploying a Machine Learning model.
- Articulate initial ethical considerations pertinent to Machine Learning applications.

TEACHING NOTES

Topic 1: What is Machine Learning?

Core concept: Machine Learning is a subset of AI that enables systems to learn from data to make predictions or decisions.

Key teaching points:

- Traditional programming vs. Machine Learning: rules defined by humans vs. rules learned from data.
- Relationship to AI and Deep Learning: AI is the broad field, ML is a part of AI, Deep Learning is a part of ML.
- Focus on pattern recognition, data-driven insights, and generalization from experience.

Real-world example: Netflix recommendation system learns your preferences from your viewing history to suggest new movies.

Discussion question: How is a modern search engine using Machine Learning different from an older, rule-based search engine?

Topic 2: Types of Machine Learning

Core concept: Machine Learning problems are categorized by the nature of the data and the learning goal.

Key teaching points:

- Supervised Learning: learns from labeled data to predict an output (e.g., classification for yes/no, regression for continuous values).
- Unsupervised Learning: finds hidden patterns or structures in unlabeled data (e.g., clustering data points, reducing dimensionality).
- Reinforcement Learning: an agent learns optimal actions through trial and error by interacting with an environment.

Real-world example: Supervised for predicting credit risk, Unsupervised for market segmentation, Reinforcement for training a robot.

Discussion question: If you wanted to automatically group similar news articles without predefined categories, which type of Machine Learning would you use?

Topic 3: Basic Machine Learning Workflow

Core concept: A standard Machine Learning project follows a sequence of steps from data preparation to model deployment.

Key teaching points:

- Data Collection and Preprocessing: acquiring relevant data, cleaning it, handling missing values, and transforming features.

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Model Training and Selection: choosing an appropriate algorithm, splitting data into training and testing sets, and f...

Model Evaluation and Deployment: assessing model performance using metrics, fine-tuning, and integrating the model in...

Real-world example: Building a face recognition system involves collecting images, labeling faces, training a neural ...

Discussion question: Why is it crucial to test a Machine Learning model on data it has not seen before during training?

TEACHING STRATEGIES

Timing breakdown:

Introduction and What is ML (15 min)

Types of ML and examples (20 min)

ML Workflow and brief ethical considerations (20 min)

Interactive activity and Q&A (15 min)

Interactive activities:

"ML or Not ML?": Present scenarios (e.g., calculator, GPS, voice assistant) and have students identify if ML is likel...

Mini-brainstorm: In small groups, ask students to identify one ML application they use daily and which ML type it lik...

Key teaching tips:

Use simple analogies and relatable examples to demystify complex concepts.

Prioritize conceptual understanding over deep technical or mathematical explanations for this introductory level.

Encourage questions frequently to ensure foundational concepts are grasped before moving on.

ASSESSMENT

Quick check questions:

What is one key difference between supervised and unsupervised learning?

Provide an example of a classification task.

Why is data cleaning an important step in the ML workflow?

Discussion prompts:

How might bias in the training data impact a Machine Learning model designed to assist with judicial sentencing?

Imagine you want to create an AI opponent for a video game that learns over time. Which type of Machine Learning woul...

Key exam topics:

Definitions of AI, Machine Learning, and Deep Learning.

Characteristics and common applications of supervised, unsupervised, and reinforcement learning.

Overview of the Machine Learning workflow steps (data preparation, model training, evaluation, deployment).

Basic understanding of ethical considerations in ML (e.g., bias, privacy).

